Uncertainty-Aware Transformer-Based Segmentation for Pelvic MRI in Endometriosis

Sehajroop Bath

Department of Electrical, Computer and Biomedical Engineering
Toronto Metropolitan University
Toronto, Canada
sehaj.bath@torontomu.ca

***Abstract*—Endometriosis segmentation from pelvic magnetic resonance imaging (MRI) remains challenging due to heterogeneous imaging protocols, structure-specific annotation grids, severe class imbalance, and limited lesion-level supervision. This work studies a Transformer-based segmentation framework for female pelvic MRI with an emphasis on reliability and uncertainty quantification. A Swin UNETR architecture with a fixed multi-modal input interface is employed, accommodating variable modality availability through sentinel-filled channels and modality-aware preprocessing. An affine-aware, per-structure label integration pipeline addresses heterogeneous annotation alignment. Training follows a two-stage strategy, consisting of organ-level pretraining and lesion-aware fine-tuning with stage-specific loss functions. Evaluation on held-out subjects shows stable learning of dominant pelvic anatomy, with uterus segmentation achieving the highest Dice scores, while ovary and endometrioma segmentation remain challenging. To complement deterministic predictions, uncertainty-aware inference using Monte Carlo dropout and test-time augmentation is applied. Qualitative analysis demonstrates that regions with poor segmentation performance consistently correspond to elevated uncertainty, indicating conservative model behavior. These results highlight uncertainty estimation as a valuable reliability signal for pelvic MRI segmentation under limited and heterogeneous supervision.**

***Index Terms*—Pelvic MRI, Endometriosis, Medical Image Segmentation, Vision Transformers, Swin UNETR, Uncertainty Quantification, Monte Carlo Dropout, Test-Time Augmentation.**

I. Introduction

Endometriosis is a chronic gynecological disorder characterized by the presence of endometrial-like tissue outside the uterine cavity, frequently involving the ovaries and surrounding pelvic structures [1]. It is associated with chronic pelvic pain, infertility, and reduced quality of life [1]. Accurate assessment of disease extent is critical for diagnosis, surgical planning, and longitudinal monitoring. Magnetic resonance imaging (MRI) is widely used for preoperative evaluation of pelvic endometriosis because it provides high soft-tissue contrast and supports localization of deep pelvic lesions and endometriomas for surgical planning [2].

Automated segmentation of pelvic organs and endometriotic lesions from MRI has the potential to reduce radiologist workload and enable reproducible quantitative disease assessment. However, this task remains challenging due to high inter-patient anatomical variability, the small size and irregular morphology of ovaries and endometriomas, and heterogeneity in MRI acquisition protocols across scanners and institutions. These difficulties are compounded by annotation ambiguity: even expert delineations can disagree on organ boundaries and lesion extent. The recently released UT-EndoMRI dataset provides multi-sequence pelvic MRI from two clinical institutions with uterus, ovary, and endometrioma labels and reports substantial inter-rater variability, particularly for ovary segmentation [3].

Convolutional neural networks (CNNs), most notably U-Net and its derivatives, have been highly successful for biomedical image segmentation [4]. The self-configuring nnU-Net framework advanced this paradigm by automatically adapting preprocessing, architecture, training, and post-processing to a given dataset, establishing a strong and widely used baseline for medical segmentation [5]. On UT-EndoMRI, nnU-Net and the RAovSeg baseline pipeline demonstrated reasonable performance for some pelvic structures but only modest performance for ovaries and endometriomas, reflecting both the intrinsic difficulty of the task and the limitations of deterministic CNN models under heterogeneous data and label ambiguity [3]. Importantly, these approaches produce point-estimate predictions without an explicit measure of confidence, which can be problematic in clinically ambiguous regions where annotation disagreement is high.

Transformer-based architectures have recently emerged as a powerful alternative to CNNs for medical image segmentation. By leveraging self-attention, Transformers can capture long-range spatial dependencies and global anatomical context that are more difficult to model with purely local convolutional operators. Hybrid and fully Transformer-based models such as TransUNet [6], UNETR [7], and Swin-UNETR [8] have demonstrated strong performance on a range of 2D and 3D medical segmentation tasks, particularly when global context is important for delineating structures with complex shape variability. In addition, self-supervised pre-training strategies for Swin Transformer backbones have shown improved downstream segmentation performance, especially in limited-label regimes [9]. Despite these advances, most Transformer-based segmentation pipelines remain deterministic and do not explicitly model predictive uncertainty.

In clinical imaging applications, uncertainty estimation is increasingly recognized as a key component of trustworthy AI systems. Predictive uncertainty can highlight regions of low confidence, identify cases likely to benefit from human review, and provide insight into model behavior under dataset heterogeneity and annotation noise. Practical uncertainty estimation methods include Monte-Carlo (MC) dropout [10], deep ensembles [11], and test-time augmentation (TTA), which can be used to estimate aleatoric uncertainty induced by plausible imaging transformations [12]. Recent work has also explored uncertainty-aware learning schemes within Transformer-based segmentation frameworks, for example by incorporating uncertainty into semi-supervised training objectives [13]. However, uncertainty-aware inference for Transformer-based pelvic MRI segmentation in endometriosis remains underexplored, particularly in the presence of multi-sequence inputs, missing modalities, and label–sequence misalignment.

This work addresses these gaps by investigating an uncertainty-aware, multi-modal Transformer framework for pelvic MRI segmentation in endometriosis using a 3D Swin-UNETR backbone [8]. The proposed approach integrates multiple MRI sequences within a unified input representation, explicitly handles missing modalities on a per-patient basis, and incorporates uncertainty estimation mechanisms during inference. A two-stage training strategy is employed, with initial pretraining on a homogeneous single-center dataset to learn robust organ features, followed by fine-tuning on a heterogeneous multi-center dataset to specialize in endometrioma segmentation [3]. Uncertainty-aware inference enables the generation of voxel-wise confidence maps that can be qualitatively assessed in relation to anatomical ambiguity and rater disagreement, supporting reliability analysis in challenging pelvic MRI cases.

The primary contributions of this work are threefold:

1) the development of a practical multi-modal Swin-UNETR pipeline for pelvic MRI segmentation that accounts for missing sequences and label–sequence misalignment;
2) a two-stage pretraining and fine-tuning strategy tailored to the characteristics of the UT-EndoMRI dataset [3]; and
3) the exploratory integration of uncertainty-aware inference (MC dropout and TTA) to support interpretability and reliability analysis in endometriosis MRI segmentation [10], [12].

Together, these contributions aim to advance automated pelvic MRI analysis toward more reliable and clinically informative segmentation systems.

II. Related Work

This section reviews prior work across three dimensions most relevant to endometriosis pelvic MRI segmentation: (i) UT-EndoMRI and established CNN-based baselines, (ii) Transformer-based medical segmentation models and pretraining strategies, and (iii) uncertainty estimation methods for dense prediction, including emerging uncertainty-aware Transformer frameworks.

A. UT-EndoMRI Dataset and Baseline Pipelines

A key barrier to progress in endometriosis MRI segmentation has been limited availability of curated, multi-sequence datasets with standardized annotations. Liang *et al.* introduced UT-EndoMRI, a multi-modal pelvic MRI dataset spanning two clinical institutions and containing expert annotations for uterus and ovaries for all subjects, with endometrioma annotations available when lesions are detectable [3]. Importantly, UT-EndoMRI includes a multi-rater subset enabling quantification of inter-rater agreement, which highlights substantial annotation ambiguity for ovaries and endometriomas [3]. This ambiguity motivates approaches that improve robustness under heterogeneity and provide mechanisms to assess predictive reliability.

Alongside the dataset release, Liang *et al.* reported baseline performance using two representative segmentation pipelines: nnU-Net and RAovSeg [3]. RAovSeg combines a slice-level classifier with an attention-based segmentation network and was proposed to improve ovary segmentation in endometriosis MRI [3]. These baselines establish strong reference points for pelvic organ segmentation performance on UT-EndoMRI, while also underscoring the persistent difficulty of reliably delineating ovaries and endometriomas under real-world multi-center variation [3].

B. CNN-Based Medical Image Segmentation Baselines

U-Net remains the foundational architecture for biomedical segmentation due to its encoder–decoder structure with skip connections that preserve spatial detail [4]. Many subsequent systems for medical segmentation extend this paradigm with residual connections, attention, and multi-scale strategies. Among these, nnU-Net is particularly influential as a self-configuring framework that adapts preprocessing, architecture, and training schedules to the dataset at hand, often yielding state-of-the-art performance without extensive manual tuning [5]. In recent comparative analyses across medical segmentation tasks, CNN-based variants such as Res-UNet and nnU-Net have demonstrated strong accuracy and recall, reinforcing the competitiveness of convolutional baselines [14]. However, CNN models are typically deterministic and may be overconfident in regions with ambiguous boundaries, which is a critical concern in endometriosis MRI where inter-rater variability is high.

C. Transformer-Based Medical Image Segmentation

Transformers were introduced to overcome locality limitations of convolution by modeling global dependencies via self-attention. In medical image segmentation, TransUNet demonstrated that combining convolutional feature extraction with a Transformer encoder can improve contextual modeling while retaining precise localization through U-Net-style decoding [6]. Extending to volumetric medical data, UNETR reformulated 3D segmentation as a sequence-to-sequence problem by applying a Transformer encoder to non-overlapping 3D patches and connecting intermediate representations to a CNN decoder via multi-scale skip connections [7]. This approach showed strong performance gains on multi-organ benchmarks, supporting the value of global context in 3D segmentation.

Swin-UNETR further improved computational efficiency by adopting hierarchical Swin Transformer encoders with shifted-window attention, enabling scalable attention while preserving cross-window context [8]. The resulting architecture combines multi-resolution Transformer features with U-Net-like decoding and has been validated on challenging 3D medical segmentation tasks, making it well suited for pelvic MRI where anatomical context across slices is important and GPU memory constraints are common [8].

Because labeled medical data is often scarce, self-supervised pre-training has emerged as a key strategy for improving representation quality. Tang *et al.* proposed self-supervised pre-training for Swin Transformer backbones using tailored proxy tasks for 3D medical image analysis, demonstrating improved downstream segmentation performance and greater robustness in low-label regimes [9]. These findings motivate transfer learning and staged training strategies for specialized domains such as endometriosis MRI segmentation.

D. Uncertainty Estimation for Segmentation

For clinical deployment, segmentation quality alone is insufficient: models must also communicate when predictions are unreliable. A widely used approach for epistemic uncertainty estimation is Monte-Carlo (MC) dropout, where dropout is retained at test time and multiple stochastic forward passes approximate a Bayesian posterior predictive distribution [10]. This enables voxel-wise uncertainty estimates (e.g., predictive entropy or variance) without requiring architectural changes, and has become a practical baseline for uncertainty-aware segmentation systems.

Complementary to MC dropout, deep ensembles estimate predictive uncertainty by aggregating outputs from multiple independently trained models, often improving calibration and robustness under distribution shift at the cost of increased computation [11]. In addition, test-time augmentation (TTA) has been formalized as a Monte-Carlo procedure over plausible image transformations and acquisition noise, supporting estimation of aleatoric uncertainty and reducing overconfident errors in medical image segmentation [12]. These methods are particularly relevant to UT-EndoMRI given multi-center heterogeneity and annotation ambiguity.

E. Uncertainty-Aware Transformers

While many Transformer segmentation pipelines remain deterministic, recent work has explored uncertainty-aware mechanisms integrated with Transformer backbones. For example, Wang *et al.* proposed an uncertainty-aware Transformer framework for MRI cardiac segmentation in a semi-supervised setting, using uncertainty to filter unreliable pseudo-labels during teacher–student consistency training [13]. Such approaches highlight a broader trend of incorporating uncertainty into training objectives and decision rules, beyond purely post-hoc uncertainty estimation.

F. Positioning of This Work

In summary, UT-EndoMRI provides a realistic benchmark for endometriosis pelvic MRI segmentation, but existing baselines (nnU-Net and RAovSeg) demonstrate that ovaries and endometriomas remain challenging under heterogeneous imaging conditions and annotation ambiguity [3]. Transformer-based models such as Swin-UNETR offer improved global context modeling and competitive performance in 3D medical segmentation [8], while uncertainty estimation techniques (MC dropout, ensembles, and TTA) provide practical mechanisms to assess reliability in ambiguous regions [10]–[12]. This motivates an approach that (i) leverages a 3D Swin-UNETR backbone for multi-sequence pelvic MRI, and (ii) integrates uncertainty-aware inference to produce voxel-wise confidence maps for interpretability and reliability analysis in endometriosis MRI segmentation.

III. Materials and Methods

Experiments were conducted using the UT-EndoMRI dataset, a publicly available multi-modal pelvic MRI dataset designed for deep learning-based segmentation in endometriosis [3]. The dataset comprises two complementary cohorts acquired at different clinical institutions. The D1_MHS cohort is multi-center and multi-rater, consisting of 51 subjects with expert annotations from up to three raters, enabling analysis of inter-rater variability. The D2_TCPW cohort is single-center and single-rater, consisting of 81 subjects with comparatively more homogeneous acquisition protocols.

Each subject includes one or more MRI sequences drawn from T1-weighted (T1), fat-suppressed T1-weighted (T1FS), T2-weighted (T2), and fat-suppressed T2-weighted (T2FS) imaging. Manual annotations are provided for uterus and ovaries for all subjects, while endometrioma annotations are available only for subjects in which lesions are visible. Importantly, modality availability is non-uniform across subjects, and annotations for different structures may be defined in different sequence spaces within the same subject.

Several dataset characteristics of UT-EndoMRI strongly influenced the design of the segmentation pipeline.

First, modality availability is non-uniform on a per-subject basis. In principle, D1_MHS includes T1FS and T2 imaging, while D2_TCPW includes T1, T1FS, T2, and T2FS; however, in practice many subjects are missing one or more sequences. As a result, the assumption of complete multi-modal inputs does not hold.

Second, structure annotations are modality-dependent. A common pattern in UT-EndoMRI is that uterus and ovary labels are aligned to a T2 volume, whereas endometrioma labels are derived from a T1FS volume. Consequently, label arrays for different structures within the same subject may have different voxel grids, shapes, spacings, and affine transforms.

Third, the dataset exhibits extreme class imbalance. Endometriomas occupy a very small fraction of voxels, are absent in many subjects, and vary substantially in size and appearance. Naïve sampling or preprocessing strategies can therefore lead to systematic under-exposure of the lesion class during training.

These constraints necessitate careful handling of missing modalities, label alignment, dataset statistics, and sampling to avoid silently discarding supervision or introducing bias.

Let $\mathbf{X} \in \mathbb{R}^{C \times H \times W \times D}$ denote a multi-channel 3D MRI volume, where $C = 4$ corresponds to the canonical modality order $\{\mathrm{T1}, \mathrm{T1FS}, \mathrm{T2}, \mathrm{T2FS}\}$. The objective is to predict a voxel-wise segmentation map $\mathbf{Y} \in \{0, \ldots, K\}^{H \times W \times D}$, where $K = 3$ during pretraining (background, uterus, ovary) and $K = 4$ during fine-tuning (background, uterus, ovary, endometrioma). The task is formulated as supervised multi-class semantic segmentation.

Swin-UNETR requires a fixed number of input channels. To accommodate heterogeneous modality availability while maintaining a consistent input interface, all subjects are represented using a fixed four-channel tensor corresponding to the canonical modality order. If a particular modality is unavailable or cannot be aligned to the reference grid, the corresponding channel is filled with a constant sentinel value (set to -1.0), and a modality mask is recorded to indicate channel validity.

This design avoids discarding subjects with incomplete modality sets and allows the network to learn modality presence and absence as a first-class condition. It enables joint training across subjects with heterogeneous modality availability while preserving consistent channel semantics across batches. Missing modalities are therefore handled explicitly rather than excluded, allowing the model to learn robustness to incomplete inputs. Variable-channel inputs or subject filtering were explicitly avoided in favor of this fixed-interface strategy.

To prevent loss of supervision due to misaligned labels, a per-subject, per-structure label-sequence alignment procedure is employed. For each anatomical structure, candidate MRI sequences available for the subject are evaluated by comparing voxel spacing, array shape, and affine transformation matrices between the label volume and each candidate image volume.

When enabled, per-structure sequence detection assigns each structure (uterus, ovary, endometrioma) to the sequence that best matches its annotation grid. A subject-level reference sequence is then selected by majority vote across structures. All label volumes are resampled into the reference image space using affine-aware nearest-neighbor interpolation to preserve categorical integrity.

This approach ensures that labels defined on different sequence grids can be merged into a single, spatially consistent multi-class label map, and avoids silently skipping labels due to shape mismatches, an issue that disproportionately affected endometrioma supervision in early pipeline iterations.

All MRI volumes are resampled to a fixed target spacing and cropped or padded to a fixed spatial size to ensure consistent tensor dimensions across subjects. Intensity normalization is applied independently per modality using percentile-based clipping followed by min-max normalization.

Given heterogeneous fields-of-view and rare foreground structures, deterministic preprocessing and label-aware cropping are prioritized. During training, random spatial crops are guided by foreground labels to ensure sufficient exposure to small structures, particularly endometriomas. Aggressive spatial augmentations are de-emphasized to avoid compounding alignment errors introduced by heterogeneous acquisition and annotation grids.

The segmentation backbone is Swin-UNETR, a hybrid Transformer–CNN architecture that combines a hierarchical Swin Transformer encoder with a convolutional decoder and U-Net-style skip connections [8]. Shifted-window self-attention enables efficient modeling of long-range spatial dependencies while maintaining scalability for 3D volumes under GPU memory constraints.

The network accepts a four-channel volumetric input and outputs voxel-wise class logits at the original resolution via a multi-scale decoding pathway.

Training is performed in two stages using a combined dataset comprising subjects from both D1_MHS and D2_TCPW, with unified splits defined over the merged subject set. In both stages, data loaders are constructed using a multi-dataset interface with per-subject modality handling and per-structure label-sequence alignment enabled.

In the first stage, the model is trained with a three-class objective (background, uterus, ovary). This stage emphasizes learning robust representations of common pelvic anatomy under heterogeneous modality availability and label alignment conditions.

In the second stage, the model is fine-tuned using a four-class objective that includes endometriomas. Fine-tuning focuses on specialization to rare lesion structures while retaining organ-level representations learned during pretraining. Sampling strategies and class weighting are adjusted to mitigate extreme class imbalance for the endometrioma class.

In both stages, training employs a composite loss consisting of Dice loss and cross-entropy loss to balance region overlap accuracy and voxel-wise classification performance.

Uncertainty estimation is incorporated at inference time using a stochastic prediction pipeline implemented in the model wrapper. Monte-Carlo (MC) dropout is enabled during inference by retaining dropout layers and performing multiple stochastic forward passes [10]. For each MC sample, test-time augmentation (TTA) is additionally applied using a set of spatial flips, yielding multiple predictions per input.

Inference is performed using sliding-window evaluation to support large volumetric inputs. For each subject, predictions are aggregated across MC samples and TTA transformations to compute a predictive mean segmentation as well as voxel-wise uncertainty measures, including normalized predictive entropy, epistemic variance (variance across MC samples), and aleatoric variance (variance across TTA transformations) [12].

Uncertainty-aware inference is treated as an exploratory analysis tool in this work. No uncertainty-specific loss terms or calibration objectives are used during training; instead, uncertainty maps are analyzed qualitatively to assess correspondence with anatomical ambiguity, missing modalities, and known inter-rater variability.

IV. Experimental Setup

A. Data Splits

All experiments use unified train/validation/test splits defined over the combined UT-EndoMRI subject set, including both D1_MHS and D2_TCPW cohorts. Subject-level splits are constructed to ensure that no subject appears in more than one split, preventing information leakage across training and evaluation. Split definitions are stored in JSON files and loaded consistently across pretraining, fine-tuning, and evaluation stages.

Because labels for different structures may reside on different sequence grids, split statistics are computed using per-structure label presence rather than merged label volumes. This avoids false negatives in lesion presence that can arise when structure labels are aligned to different voxel grids, particularly for endometriomas. As a result, endometrioma-positive subjects are preserved during split generation and sampling.

B. Preprocessing Configuration

All MRI volumes are resampled to a fixed target spacing and cropped or padded to a fixed spatial size prior to training and inference. This ensures consistent tensor dimensions across subjects with heterogeneous acquisition protocols. Intensity normalization is applied independently per modality using percentile-based clipping followed by min-max normalization.

A deterministic preprocessing pipeline is employed to minimize confounding effects from geometric inconsistencies and to ensure stable label alignment. Spatial preprocessing is applied identically across modalities for a given subject to preserve voxel-wise correspondence.

C. Data Augmentation

During training, data augmentation is applied conservatively to improve generalization while avoiding compounding alignment errors. Augmentation strategies include random flips and label-aware random cropping to ensure that foreground structures are present within training patches. Label-aware cropping is particularly important given the small size and sparse distribution of endometriomas.

Aggressive spatial deformations and intensity perturbations are de-emphasized due to heterogeneous acquisition geometry and the presence of structure-specific label grids, which can make such augmentations destabilizing.

D. Loss Functions and Class Imbalance Handling

Training employs stage-specific loss functions designed to address both region overlap accuracy and severe class imbalance, particularly for endometrioma segmentation.

1) Stage 1: Pretraining Loss: During Stage 1 pretraining, the model is optimized using a composite Dice-Cross-Entropy loss. Specifically, Dice loss with softmax activation and one-hot targets is combined with weighted categorical cross-entropy:

$$\mathcal{L}_{\text{pre}} = 0.7\,\mathcal{L}_{\text{Dice}} + 0.3\,\mathcal{L}_{\text{CE}}. \tag{1}$$

Dice loss encourages spatial overlap between predictions and ground truth, while cross-entropy promotes voxel-wise classification accuracy. Class weights are applied in the cross-entropy term to partially compensate for foreground–background imbalance. This loss configuration is used for the three-class pretraining objective (background, uterus, ovary), where anatomical structures are relatively stable and consistently annotated.

2) Stage 2: Fine-Tuning Loss: During Stage 2 fine-tuning, the loss function is modified to better handle extreme class imbalance introduced by the endometrioma class. A focal loss formulation is adopted to down-weight easy examples and emphasize hard-to-classify voxels:

$$\mathcal{L}_{\text{fine}} = 0.7\,\mathcal{L}_{\text{Dice-Focal}} + 0.3\,\mathcal{L}_{\text{Focal}}. \tag{2}$$

Here, Dice-Focal loss combines Dice loss with focal modulation, while voxel-wise focal loss further concentrates learning on misclassified and low-confidence regions. Both terms use softmax activation and one-hot targets, with class weights applied to mitigate imbalance. A focal parameter γ controls the strength of down-weighting for easy examples and is fixed across experiments.

This stage-specific loss design reflects the progression from learning robust organ-level anatomy during pretraining to emphasizing rare lesion structures during fine-tuning.

Extreme class imbalance is addressed through a combination of loss design, class weighting, and sampling strategies. While Dice-based terms mitigate imbalance by construction, focal modulation further reduces dominance of background voxels during fine-tuning. In addition, class weights are computed from training data statistics and applied consistently within cross-entropy and focal loss components.

Together, these mechanisms ensure that rare endometrioma voxels contribute meaningfully to the optimization process without destabilizing training.

E. Optimization and Training Details

Models are optimized using the AdamW optimizer. A shared optimization framework is used across training stages, with stage-specific hyperparameters.

During Stage 1 pretraining, the learning rate is set to 2×10^{-4} to facilitate efficient learning of common anatomical structures. During Stage 2 fine-tuning, the learning rate is reduced to 5×10^{-5} and weight decay to 1×10^{-6} to enable stable optimization when learning the rare endometrioma class.

Gradient clipping with a maximum ℓ_2 norm of 1.0 is applied at every training step to improve optimization stability. Training is performed using mixed-precision arithmetic to reduce GPU memory consumption and accelerate computation. Early stopping based on validation performance is employed to mitigate overfitting, and model checkpoints are saved periodically throughout training to enable reproducibility and post hoc analysis. Training and evaluation are performed on one *NVIDIA A100* GPU with 40 gigabyte memory to support 3D sliding-window inference.

TABLE I
TEST PERFORMANCE. DICE IS REPORTED PER CLASS AND AS MEAN FOREGROUND DICE (EXCLUDING BACKGROUND).
ut : $uterus$; ov : $ovary$; em : $endometrioma$

Stage	Best Epoch	Mean FG Dice	ut Dice	ov Dice	em Dice
PT (3-class)	25	0.19	0.38	0.00	—
FT (4-class)	24	0.11	0.34	0.00	0.00

TABLE II
EVALUATION SUMMARY. ut: UTERUS; ov: OVARY; em: ENDOMETRIOMA.

Stage	Epoch	Prio. Cls.	Prio. Dice	Mean Dice
PT	25	ut+ov	0.19	0.13
FT	24	ov+em	0.00	0.09

F. Inference & Evaluation Setup

Inference is performed using sliding-window evaluation to support full-resolution 3D volumes that exceed GPU memory limits. Overlapping windows are aggregated using weighted averaging to reduce boundary artifacts. This procedure is applied consistently for deterministic and uncertainty-aware inference. For uncertainty-aware inference, Monte-Carlo dropout is enabled and multiple stochastic forward passes are performed. For each stochastic pass, test-time augmentation via spatial flips is applied, yielding multiple predictions per subject. Predictions are aggregated to compute a predictive mean segmentation and voxel-wise uncertainty maps.

Segmentation performance is evaluated using standard overlap-based metrics, including the Dice similarity coefficient (DSC) and Intersection over Union (IoU), computed separately for uterus, ovary, and endometrioma classes. Metrics are reported per class and averaged across subjects in the test set. Uncertainty estimates are evaluated qualitatively through visualization of predictive entropy and variance maps overlaid on MRI slices. Quantitative uncertainty calibration metrics are not the primary focus of this work and are treated as exploratory.

All experiments are implemented using PyTorch and the MONAI framework. Training, inference, and uncertainty estimation are executed through a unified codebase, with configuration files and notebooks controlling experimental parameters. Experiments are run with fixed random seeds to promote reproducibility.

V. RESULTS

This section summarizes the quantitative performance and training behavior of the proposed framework across the two training stages: (i) organ-level pretraining and (ii) lesion-aware fine-tuning. Results are reported using Dice similarity coefficients computed on held-out validation and test subjects.

A. Stage 1: Organ-Level Pretraining

During Stage 1 pretraining, the model was trained using a three-class objective (background, uterus, ovary). As shown in Fig. 1, the optimization process was stable, with training loss decreasing monotonically from approximately 0.81 to 0.49 over 35 epochs. Validation Dice improved rapidly during early epochs and gradually plateaued, indicating effective learning of global pelvic anatomy.

The best validation checkpoint was obtained at epoch 25. At this point, the mean foreground Dice on the test set reached 0.19 (Table I), with uterus segmentation achieving the highest per-class Dice (0.38), while ovary segmentation remained substantially more challenging. These results confirm that the model learned stable representations of dominant pelvic structures while struggling with smaller and more variable anatomy.

B. Stage 2: Lesion-Aware Fine-Tuning

Stage 2 fine-tuning introduced a four-class objective by adding endometrioma as a target structure. As shown in Fig. 1, training loss again decreased smoothly, from approximately 0.92 to 0.53 over 44 epochs, demonstrating stable optimization under the more challenging loss formulation.

Validation mean Dice increased from 0.22 in early epochs to approximately 0.33 and remained stable thereafter. However, priority Dice (ovary + endometrioma) remained extremely low throughout training (Fig. 1). Per-class evaluation in Table I revealed that while uterus segmentation performance was retained (Dice $\approx$ 0.34), ovary Dice and and endometrioma Dice was effectively zero on the test set.

This behavior reflects the extreme class imbalance and label sparsity present in the fine-tuning data, as well as residual misalignment and modality heterogeneity across subjects. Importantly, the absence of performance gains on endometrioma segmentation does not correspond to training instability or divergence; rather, the smooth loss curves and stable validation Dice observed in Fig. 1 suggest that the model converged to a conservative solution dominated by organ-level predictions.

Across both stages, training exhibited consistent and stable dynamics (Fig. 1). No loss explosions or validation collapses were observed, and early convergence was followed by long plateaus rather than oscillatory behavior. This suggests that the optimizer configuration, gradient clipping, and loss formulations were effective in preventing instability despite heterogeneous inputs and severe class imbalance. Notably, pretraining produced smoother and higher validation Dice trajectories compared to fine-tuning, underscoring the importance of robust anatomical initialization prior to lesion-specific learning.

Although fine-tuning did not yield strong deterministic segmentation performance for endometriomas (Tables I and II), uncertainty-aware inference was successfully executed for both stages. For each best checkpoint, Monte Carlo dropout and test-time augmentation produced voxel-wise predictive entropy and variance maps, which were saved for qualitative analysis. The observed discrepancy between stable training metrics (Fig. 1) and poor lesion Dice reinforces the motivation for uncertainty estimation as a complementary signal, particularly in regions with ambiguous boundaries, missing modalities, or sparse labels. Rather than over-interpreting low Dice scores,

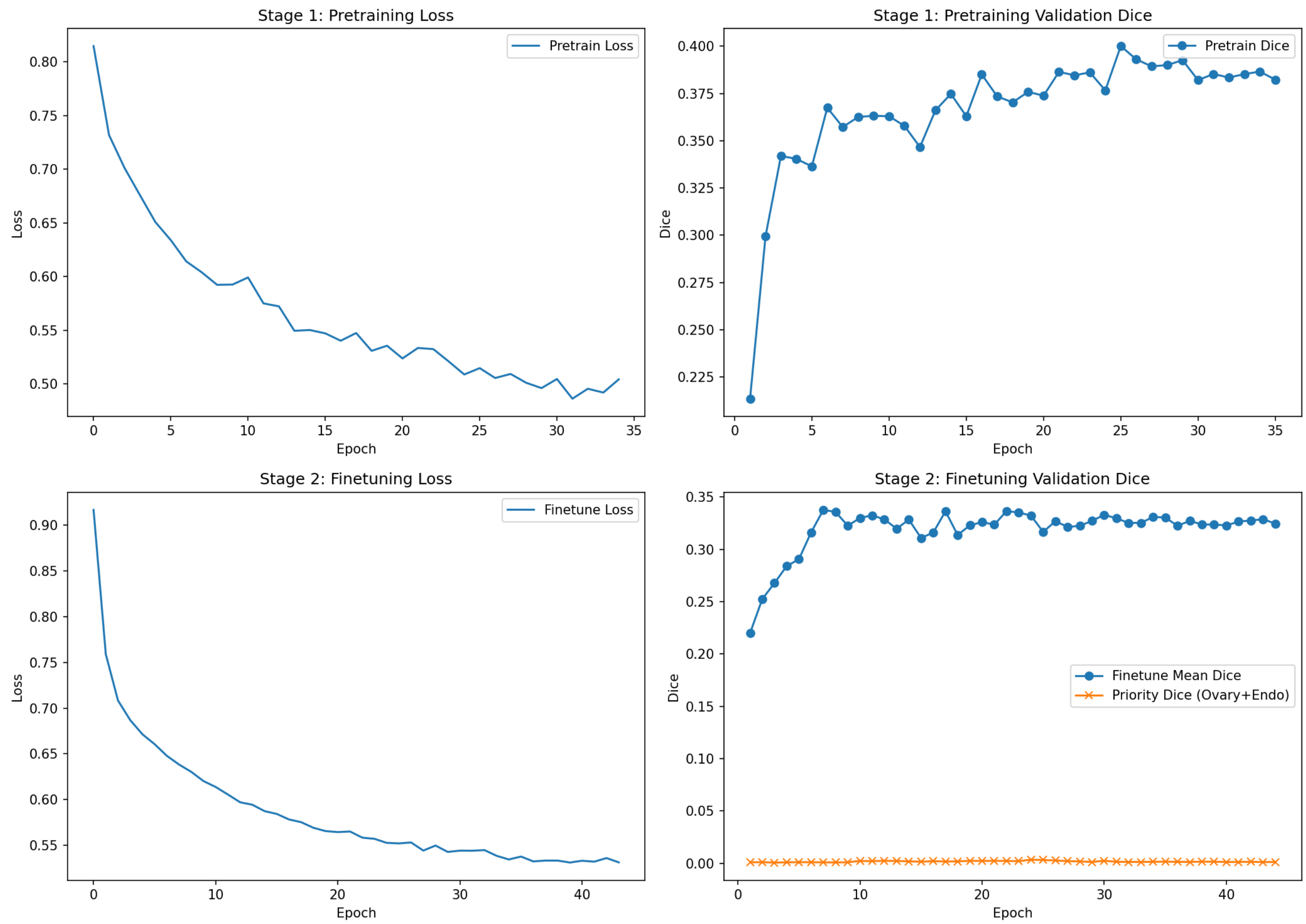

Fig. 1. Training dynamics for Stage 1 (pretraining) and Stage 2 (fine-tuning). Left: training loss. Right: validation Dice. For fine-tuning, the mean validation Dice and the priority Dice (ovary + endometrioma) are shown.

the framework emphasizes calibrated uncertainty as a tool for downstream clinical decision support.

VI. Uncertainty and Reliability Analysis

Representative uncertainty maps for the pretraining and fine-tuning checkpoints are shown in Fig. 2 and Fig. 3, respectively. In both figures, subjects are ordered from top to bottom as D2-057, D2-065 (ovary-only cases), followed by D1-001 and D1-021 (ovary and endometrioma cases). Each row shows normalized predictive entropy, epistemic uncertainty, and aleatoric uncertainty for a representative axial slice.

Uncertainty-aware inference was performed using Monte Carlo (MC) dropout combined with flip-based test-time augmentation (TTA). For each subject, multiple stochastic forward passes were aggregated to compute voxel-wise predictive entropy, epistemic variance (across MC samples), and aleatoric variance (across TTA transformations). All uncertainty maps were evaluated qualitatively in anatomically relevant regions and interpreted in relation to deterministic segmentation performance.

A. Ovary-Only Cases

For ovary-only subjects (D2-057 and D2-065), pretraining uncertainty maps in Fig. 2 exhibit structured and anatomically meaningful patterns. Predictive entropy is concentrated primarily along organ boundaries and within the pelvic cavity, while remaining relatively low in homogeneous background regions. This behavior indicates uncertainty driven by boundary ambiguity rather than random noise. Epistemic uncertainty in these cases is low and spatially sparse, with localized responses near complex anatomical interfaces, suggesting that the pretrained model has learned stable representations for dominant pelvic anatomy. In contrast, aleatoric uncertainty is more pronounced near soft-tissue interfaces and regions affected by signal variability, reflecting sensitivity to acquisition-related noise and modality-dependent appearance.

After fine-tuning (Fig. 3), uncertainty patterns for ovary-only subjects change noticeably. Predictive entropy becomes more spatially uniform across the field of view, and boundary-focused structure is reduced. Both epistemic and aleatoric uncertainty increase diffusely, particularly in regions associated with missing or inconsistent modality information. This shift suggests that fine-tuning under heterogeneous supervision introduces additional uncertainty even in anatomically consistent cases when lesion-specific signals are sparse or absent.

B. Endometrioma-Positive Cases

For subjects with annotated endometriomas (D1-001 and D1-021), uncertainty behavior differs substantially from ovary-only cases. During pretraining (bottom rows of Fig. 2), predictive entropy is elevated throughout the pelvic region, with pronounced responses near the ovaries and suspected lesion locations despite the absence of an explicit endometrioma class in the pretraining objective. Epistemic uncertainty in these cases is spatially structured and concentrated around ambiguous regions, indicating model uncertainty driven by insufficient task-relevant supervision. Aleatoric uncertainty is substantially higher than in ovary-only subjects, particularly near lesion-adjacent regions and along tissue interfaces, suggesting that small spatial perturbations introduced by TTA produce large variations in predicted probabilities. This behavior is consistent with the heterogeneous appearance of endometriomas across sequences and subjects.

Following fine-tuning (Fig. 3), predictive entropy remains elevated in lesion regions even though deterministic endometrioma Dice remains near zero. Epistemic uncertainty becomes more localized but increases in magnitude, indicating that the model explicitly represents its lack of confidence in lesion-specific predictions rather than producing overconfident false positives. Aleatoric uncertainty remains high and spatially diffuse, reflecting sensitivity to imaging variability and residual label-sequence misalignment.

Across all evaluated subjects, regions associated with higher Dice scores (e.g., uterus during both stages) correspond to lower uncertainty, while regions with poor segmentation performance (ovaries and endometriomas) consistently exhibit elevated predictive entropy and variance. Importantly, segmentation failures in lesion regions are accompanied by increased epistemic and aleatoric uncertainty rather than confident misclassification.

This conservative failure mode is particularly evident in endometrioma-positive subjects, where deterministic segmentation fails but uncertainty maps in Fig. 2 and Fig. 3 consistently highlight the same regions as unreliable.

The uncertainty maps presented in Fig. 2 and Fig. 3 demonstrate that uncertainty-aware inference provides spatially resolved indicators of prediction reliability that are not captured by Dice metrics alone. In challenging cases involving missing modalities, heterogeneous label grids, or rare lesions, uncertainty estimation enables the identification of regions that warrant human review. While uncertainty estimation does not compensate for limited lesion segmentation performance, it supports safer interpretation of model outputs by explicitly representing confidence. As such, uncertainty-aware inference constitutes a practical and valuable complement to deterministic segmentation for pelvic MRI analysis in endometriosis.

VII. Discussion

This work investigated Transformer-based pelvic MRI segmentation for endometriosis under realistic clinical constraints, including heterogeneous modality availability, structure-specific label grids, and severe class imbalance. While deterministic segmentation performance for endometriomas remained limited, the results provide several important insights into model behavior, reliability, and the role of uncertainty-aware inference in low-supervision medical imaging settings.

Quantitative results demonstrate that the proposed framework learns stable representations for dominant pelvic anatomy, particularly the uterus, during both pretraining and fine-tuning. This is reflected in consistently higher Dice scores for uterine segmentation and smooth, well-behaved training dynamics. In contrast, ovary segmentation remained challenging across both stages, and endometrioma segmentation performance was near zero despite explicit inclusion during fine-tuning. These findings are consistent with prior reports on UT-EndoMRI, which highlight substantial inter-rater variability and annotation ambiguity for ovaries and endometriomas. The limited lesion performance should therefore be interpreted in the context of extreme voxel-level class imbalance, sparse lesion annotations, and heterogeneous label-sequence alignment rather than as a failure of optimization or model capacity.

The results underscore the significant impact of dataset heterogeneity on segmentation performance. Missing modalities, inconsistent fields-of-view, and structure-specific annotation grids introduce uncertainty that cannot be fully resolved through architectural improvements alone. While the fixed-channel multi-modal input representation and affine-aware label alignment prevented silent loss of supervision, these measures primarily stabilize training rather than guarantee improved lesion Dice.

Fine-tuning on a combined multi-center dataset exposed the model to broader variability but also introduced conflicting supervision signals, particularly for rare lesion structures. The observed degradation in ovary Dice during fine-tuning highlights the trade-off between specialization for rare classes and preservation of previously learned anatomy.

A key contribution of this work is the integration and analysis of uncertainty-aware inference using Monte Carlo dropout and test-time augmentation. Although uncertainty estimation did not improve deterministic segmentation metrics, uncertainty maps provided spatially meaningful indicators of prediction reliability. In endometrioma-positive subjects, regions where deterministic segmentation failed were consistently associated with elevated predictive entropy and variance. This conservative failure mode, characterized by high uncertainty rather than confident misclassification, is particularly important in clinical applications, where overconfident false positives or false negatives may be misleading.

The alignment between high uncertainty and known sources of ambiguity, such as missing modalities and inconsistent lesion appearance, supports the use of uncertainty estimation as a complementary signal for reliability assessment rather than as a substitute for improved supervision.

A. Limitations

Several limitations should be noted. First, the dataset size and lesion prevalence limit the statistical power of quantitative evaluation for endometrioma segmentation. Second, uncertainty estimation was applied exclusively at inference time and was not incorporated into the training objective or sampling strategy, due to the compute constraints. As a result, uncertainty signals were descriptive rather than corrective.

Third, evaluation focused primarily on overlap-based metrics and qualitative uncertainty analysis. Additional calibration-focused metrics and clinician-in-the-loop evaluation would be necessary to fully assess clinical utility.

B. Implications and Future Work

The findings suggest several directions for future research. Incorporating uncertainty-aware objectives into training, such as uncertainty-weighted losses or consistency-based regularization, may help leverage uncertainty signals more effectively. Improved handling of multi-rater annotations, including probabilistic label fusion or soft labels, could further reduce ambiguity in lesion supervision. From a systems perspective, uncertainty-aware segmentation frameworks may be most valuable when integrated into semi-automated clinical workflows, where uncertainty maps guide targeted human review rather than replacing expert judgment.

An additional promising direction is large-scale pretraining on general female pelvic anatomy datasets prior to lesion-specific fine-tuning. Public resources such as the University of Maryland (UMD) pelvic MRI datasets and related multi-organ female anatomy collections provide broader anatomical coverage and more consistent organ-level supervision than disease-focused cohorts. [15] Pretraining Transformer-based models on such datasets could enable the learning of robust, modality-invariant representations of pelvic anatomy, which may then be adapted via fine-tuning to endometriosis-specific lesion detection tasks. This curriculum-style strategy: general anatomy pretraining followed by pathology-focused specialization, may mitigate data scarcity and label ambiguity while improving sample efficiency and reliability for rare lesion segmentation.

Overall, this work positions uncertainty-aware Transformer-based segmentation as a reliability-oriented approach for challenging pelvic MRI tasks, emphasizing transparent model behavior under uncertainty rather than overconfident predictions in low-supervision regimes.

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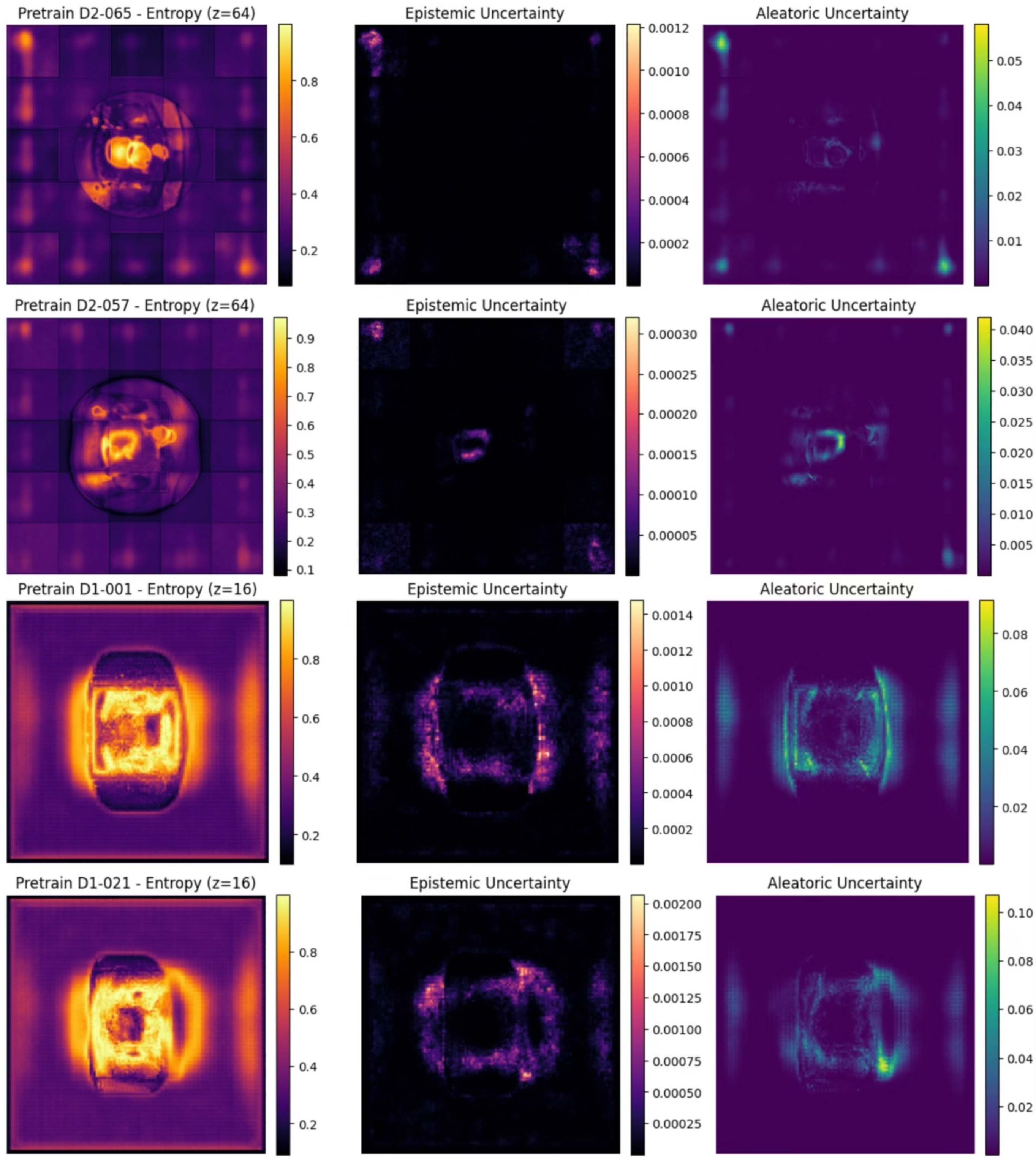

Fig. 2. Uncertainty maps for the pretraining checkpoint. Rows (top to bottom): D2-057, D2-065 (ovary-only), D1-001, D1-021 (ovary+endometrioma). Columns (left to right): predictive entropy, epistemic uncertainty, aleatoric uncertainty.

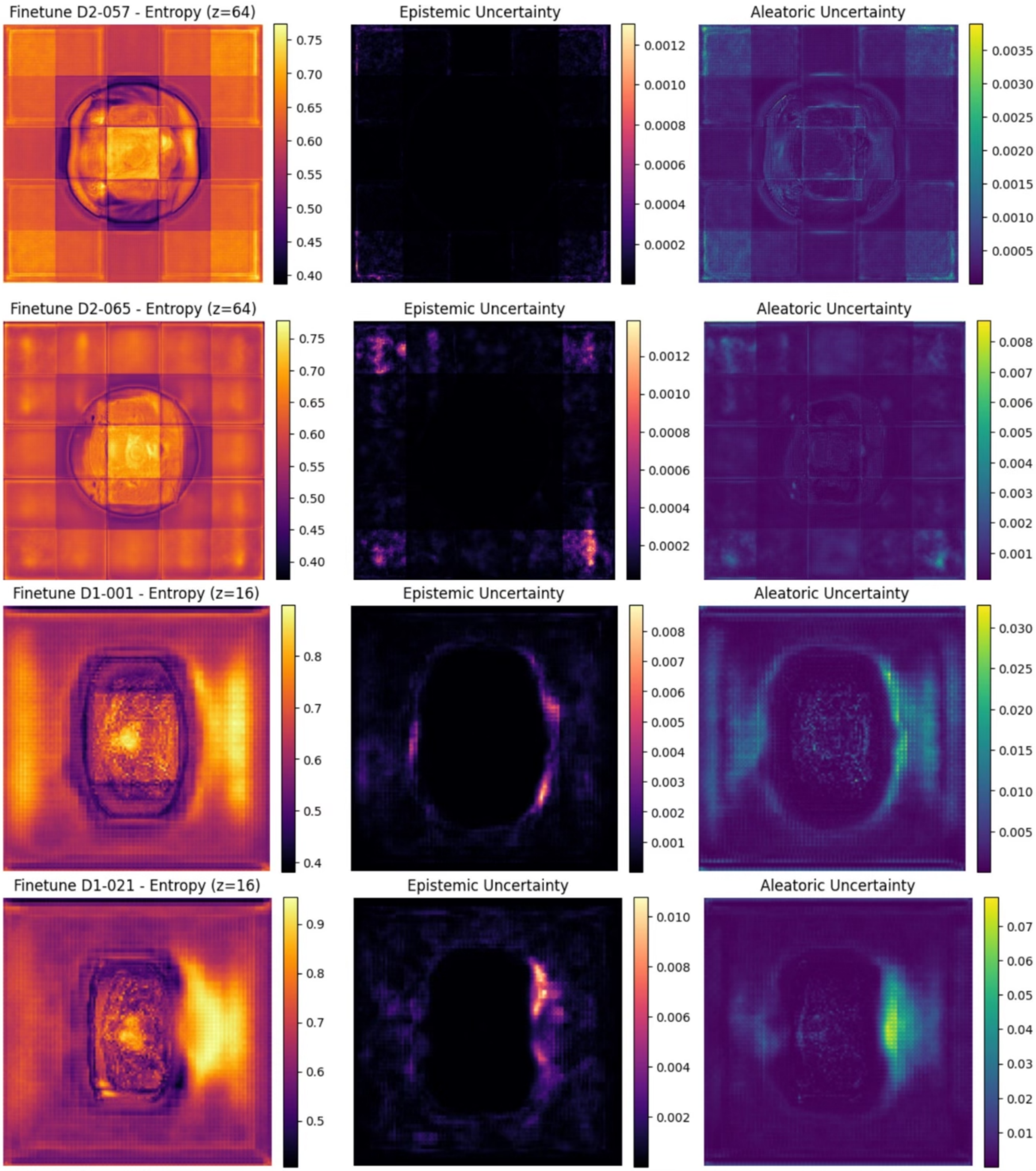

Fig. 3. Uncertainty maps for the fine-tuning checkpoint. Rows (top to bottom): D2-057, D2-065 (ovary-only), D1-001, D1-021 (ovary+endometrioma). Columns (left to right): predictive entropy, epistemic uncertainty, aleatoric uncertainty.